Beyond Discover Weekly: Building a Content-Based Spotify Discovery Engine

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GitHub

The Problem

Spotify's Discover Weekly playlist relies heavily on collaborative filtering, creating popularity bias. Since Spotify recommends what other users listen to, it misses the mathematical DNA of the actual song.

Objective

Build a content-based recommendation engine that ignores popularity and user history. Find the distinct musical clusters using machine learning to output songs with mathematically similar DNA and mood.

Methodology

Gathered 114,002 tracks from Kaggle and RapidAPI.

Extracted audio features, engineered custom feature (energy_valence_ratio), and scaled.

Evaluated the optimal number of K-Means clusters (K=5) using Silhouette and Elbow methods and used these scores to prove the Autoencoder (Non-linear) outperformed PCA (Linear) at creating better latent spaces.

Created a user centroid with their songs to apply Cosine Similarity and return the most recommended song based on song's DNA.

One of the inputs was "Magic Spells" by Crystal Castles and the output was "Modul 1" by Arovane.

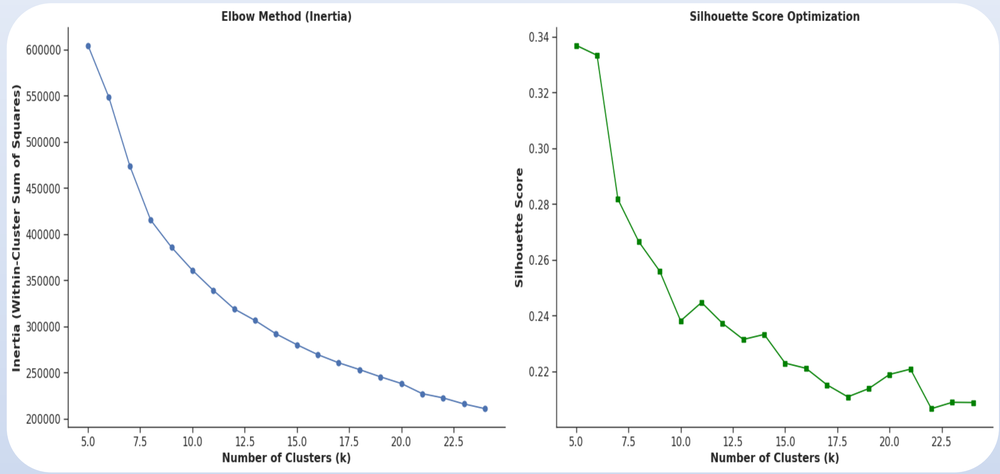

Figure 1: Silhouette evaluation defining the optimal number of clusters.

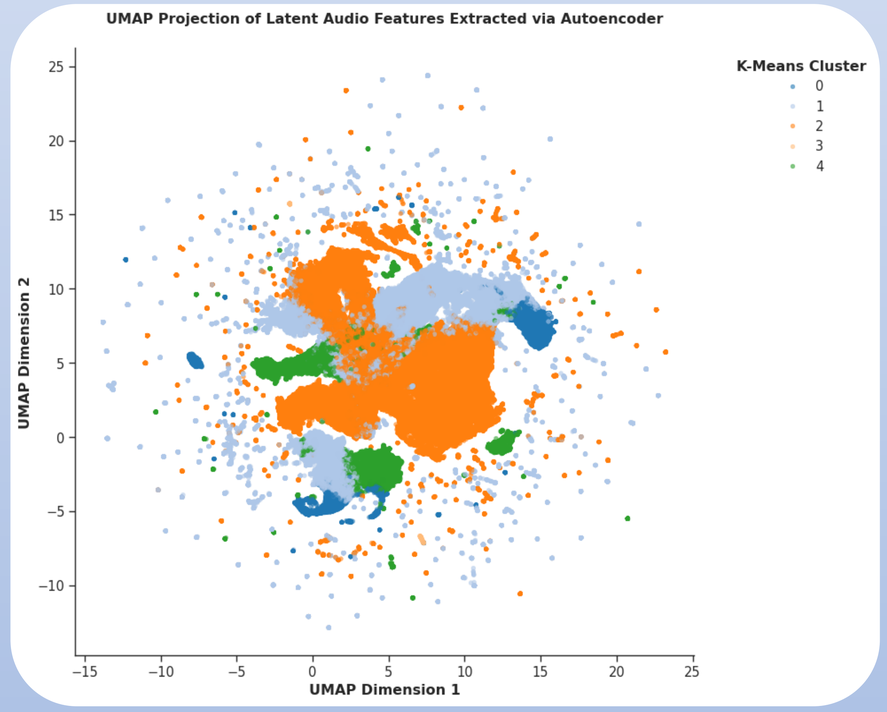

Figure 2: UMAP projection of the Autoencoder latent space colored by the K-Means cluster labels.

Key Findings

The Autoencoder outperformed the PCA, achieving a Silhouette Score of 0.277 versus 0.166. The neural network was successful and mapped complex, nonlinear relationships that linear techniques could not.

The Kaggle data set organizes best into exactly 5 distinct musical clusters.

By averaging a listener's music taste, we created a single centroid for them. We then used this centroid to recommend highly similar tracks based entirely on their audio features.

Future Considerations

Don't drop key, mode, and major.

Create a front-end to allow other users to input their own songs.

API to add songs to personal playlist.

References

[1] Maharshi Pandya. (2022). Spotify Tracks Dataset [Data set]. Kaggle. https://doi.org/10.34740/KAGGLE/DSV/4372070

[2] SoundNet. (n.d.). Track Analysis API. RapidAPI. Retrieved April 21, 2026, from https://rapidapi.com/soundnet-soundnet-default/api/track-analysis